Bihar Engineering University, Patna

B.Tech 5th Semester Examination, 2024

Course: B.Tech | *Time: 02 Hours*

Code: 101506 | *Subject: Mechanics of Materials* | *Full Marks: 20*

Instructions:-

(i) The marks are indicated in the right-hand margin

*(ii) There are **NINE** questions in this paper.*

*(iii) Attempt **FIVE** questions in all.*

(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer from the following (Any seven question only):*** **[2 x 7 = 14]**

(a) In dam analysis, the main cause of instability is due to
- (i) Overturning moment
- (ii) Vertical stress
- (iii) Axial tension
- (iv) Shear stress

(b) Cantilever retaining wall is designed mainly to resist _____ form backfill
- (i) Active earth pressure
- (ii) Water pressure
- (iii) Passive earth pressure
- (iv) Uplift earth pressure

(c) Hill's yield criterion is used for
- (i) Brittle materials
- (ii) Ductile materials
- (iii) Anisotropic materials
- (iv) Orthotropic materials

(d) According to Von Mises, yield locus is a/an
- (i) Circle
- (ii) Hexagon
- (iii) Rectangle
- (iv) Ellipse

(e) The compatibility conditions are essentially required to solve
- (i) Substitute frame
- (ii) Compound truss
- (iii) Complex truss
- (iv) Redundant frame

(f) If the member is denoted by m and joints by j, the condition for a frame to be redundant is given by
- (i) $m < 2j + 3$
- (ii) $m > 2j + 3$
- (iii) $m < 2j - 3$
- (iv) $m > 2j - 3$

(g) In multiaxial stress-strain relations, volumetric strain is associated with
- (i) Shear strain
- (ii) Yield stress
- (iii) Hydrostatic stress
- (iv) Deviatoric stress

(h) According to Castigliano's first theorem, partial derivative of strain energy with respect to force gives
- (i) Normal force
- (ii) Shear force
- (iii) Bending moment
- (iv) Deflection

(i) The virtual work method is used to determine
- (i) Beam slope and deflection
- (ii) Bending moment
- (iii) Plastic collapse load
- (iv) Principal stress

(j) In yield design using limit analysis, which principle is applied?
- (i) Virtual displacement
- (ii) Lower and upper bound theorems
- (iii) Compatibility
- (iv) Maxwell's reciprocity

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Q.2 (a) Describe how the stability of a gravity dam is analyzed. What are the different types of failures a dam structure can undergo? [7]

(b) A circular chimney of brick masonry has uniform section throughout its height, having an inner diameter of 2 m and outer diameter of 3 m. find the maximum permissible height of the chimney, if no tension is to develop and if the allowable compressive stress in masonry is 1.5 N/mm^2. Take a uniform wind pressure intensity of 1.5 kN/mm^2 and unit weight of brick masonry as 22 kN/mm^3. [7]

Q.3 (a) Explain the physical meaning of linear strain, shear strain and volumetric strain. Discuss how these differ when using finite vs infinitesimal strain theory. [7]

(b) Differentiate between Tresca and Von-Mises yield criteria. Discuss their applicability for ductile materials. [7]

Q.4 (a) Explain how momentum balance is applied to slender structural members. What are the assumptions involved? [7]

(b) A simply supported beam of span 10 m carry as UDL of 10 kN/m over a length of 3 m from left support and also from right support. Draw SF and BM diagram. [7]

Q.5 (a) Explain the concept of static indeterminacy. How is it determined for beams and trusses? Provide examples. [7]

(b) Discuss the method of joints and method of sections used for the analysis of trusses. When is each method preferable? [7]

Q.6 (a) In a triangular truss, the base is 4 m and height is 3 m. A vertical load of 12 kN is applied at the top joint. Determine the force in each member using method of joints. [7]

(b) Describe how Poisson's effect manifests in multiaxial stress situations. How is volumetric strain affected? [7]

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Q.7 (a) Define the terms elastic limit, proportional limit, and yield point. Discuss the relevance of these limits in the context of designing thin-walled pressure vessels. [7]

(b) A thin cylinder of 100 mm internal diameter and wall thickness 2 mm has its ends closed by rigid plates and is then filled with water. When an external pull of 20 kN is applied to the ends, the water pressure read by the gauge, is observed to fall by 0.075 N/mm^2 neglecting any end effect due to plates, determine the value of Poisson's ratio for the metal. Take E for metal = 2.1×10^5 N/mm^2, and bulk modulus of water = 2.17×10^3 N/mm^2 [7]

Q.8 (a) A straight bar of mild steel, 1 m long and 12mm x 6mm in section is mounted in strut testing machine and loaded axially till it buckles. Assuming Euler's formula for pinned ends to apply, estimate the maximum central deflection before the material attains its yield point of 300 N/mm^2. Take E= 2×10^5 N/mm^2. [7]

(b) Derive multi-axial stress-strain relationship among shear stresses and strains for linear isotropic elastic materials. [7]

Q.9 ***Write Short notes on any two of the following:*** [7 x 2 = 14]

(a) Hill's yield criteria and physical significance of its coefficient.

(b) Lame's theory and its assumption.

(c) Types of load, support and beams.

(d) Euler's theory, its assumption and limitation.

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